

Rachel Lin

Cheshire, CT | rroe1825@gmail.com | (802)-324-8226 | [Github](#) | [LinkedIn](#) | [Portfolio](#)

EDUCATION/AWARDS

Amherst College | Amherst, MA

May 2025

- Bachelor of Arts: Computer Science
- Relevant Coursework: Computing and the Arts, Statistics via Modeling, Linear Algebra via Applications, Data Structures, Algorithms, Artificial Intelligence, Data Science, Machine Learning, Data Mining

GPA: 3.74/4

Major League Hacking CivicHacks Hackathon | Third Place, Best Accessibility Hack

January 2023

CodePath Technical Programs | Selected for Advanced Technical Interview Prep + Applications of AI Engineering

June 2026

EXPERIENCE

Full Stack Developer (Volunteer) @ Keelung For a Walk Program / Keelung, Taiwan

February 2026 - Present

- Developing admin and user-facing features in a Vue.js SPA to manage and configure interactive hunt experiences, including WebAR challenges (AR.js), geolocation-based logic, and multilingual map support
- Integrating client-side state persistence and deployment workflows via Firebase, supporting scalable delivery of 200+ unique challenge codes

Full Stack Developer (Contract) @ IHAI Time (Initiative) | Remote

July 2025 - Present

- Developing backend services, REST APIs, and relational data architecture using FastAPI and PostgreSQL for a React-based volunteer management platform; built AI-assisted ingestion workflows using prompt-engineered LLMs to standardize spreadsheet data and automate token-credit issuance
- Collaborating with cross-functional teams to establish partnerships with 3+ nonprofits and student organizations, supporting pilot rollout and early user adoption

Data Mining Intern | Amherst College | Amherst, MA

January 2025 - May 2025

- Built a tweet data pipeline using *Selenium* to collect and analyze 10,000+ tweets; applied NLP models in machine learning (via sentiment analysis) to uncover trends across Latin American feminist hashtags
- Developed 15+ visualizations using *pandas* + *seaborn* libraries to communicate engagement patterns and temporal trends, supporting data-driven insights into hashtag performance

COSC-112 Teaching Assistant | Amherst College | Amherst, MA

September 2023 - December 2023

- Led weekly lab sessions and 12+ office hours for ~40 students in Intro to Computer Science II, supporting projects and coursework in core artificial intelligence concepts and data structures and algorithms

Research Fellowship → Gregory Call Academic Intern | Amherst College | Amherst, MA

June 2023 - December 2023

- Conducted research on queuing theory models, simulating pass-and-swap queues in Python to analyze system performance under varying load conditions
- Modeled a Jackson network (M/M/k) to study customer behavior (joining vs. balking) and evaluate throughput; applied statistical modeling to validate findings

PROJECTS

Formula 1 Data Blog Project | Python, R, Plotly, Leaflet

January 2025

- Built an end-to-end ETL pipeline to ingest, clean, and transform historical Formula 1 race data from multiple public datasets into interactive visualizations and in-depth analyses on a blog website deployed on Github Pages
- Developed interactive analytics as visualizations in R using Plotly and geospatial tools (*Leaflet*) to visualize race performance, geographic trends, and constructor statistics through dynamic charts and maps.

Anti-Brainrot | Electron (IPC), React/Vite, JS, TensorFlow.js (COCO-SSD), Node.js | [Link](#)

April 2026

- Built an Electron + React desktop app using TensorFlow.js to detect phone usage in real time (~100ms inference), improving detection accuracy through custom false-positive mitigation logic
- Architected a multi-window IPC system for behavioral intervention overlays and iteratively refining reliability through informal testing with ~10 users

OTHER/ACTIVITIES

The Remedy Project (Civic Advocacy Organization) | Remote

January 2023 - February 2024

Lead Data & IT Team Member (Volunteer)

- Managed backend operations of the action-page platform using NationBuilder, supporting supporter data workflows, CRM segmentation, donation pipelines, and campaign automation for 1,000+ community members
- Collaborated with co-founders to improve website architecture and user experience by customizing templates with Jinja, HTML, CSS, and JavaScript, building dynamic landing pages and streamlining action-page conversions

SKILLS & INTERESTS

LANGUAGES: Python, Javascript, Java, HTML5/CSS3, JS, SQL (Postgres), Typescript, R

FRAMEWORKS: React.js, Node.js, Vue.js

SKILLS & TOOLS: Github/Git, VSCode, RStudio, Vercel, Xcode, Excel